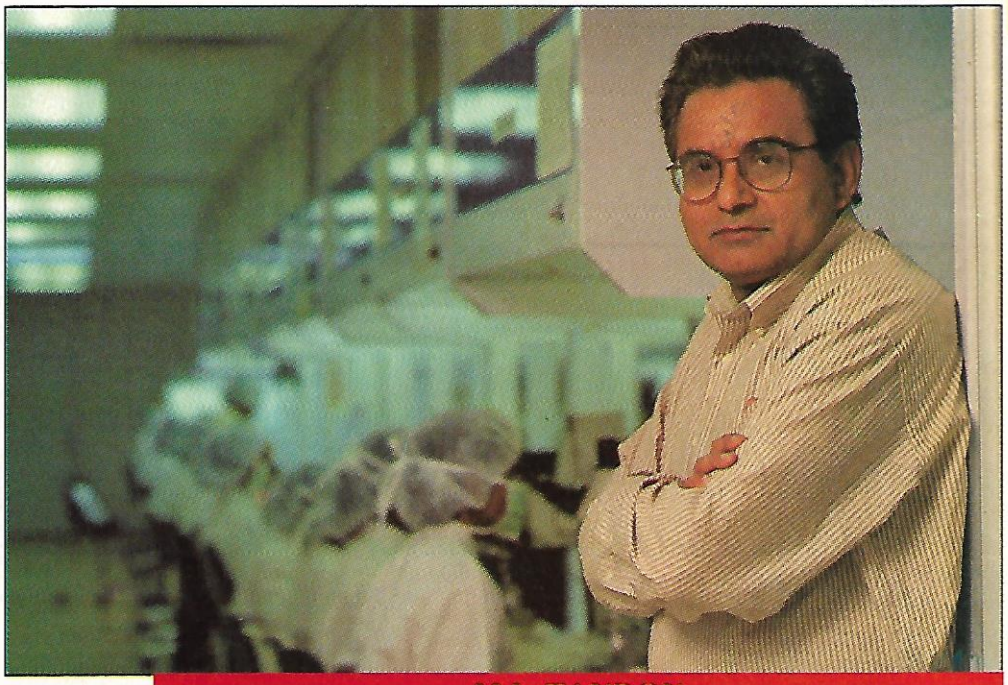
developed markets demand. And it is working: HCL-HP's exports have shot up from Rs 9 crore to Rs 40 crore in 1993-94.

Another would-be global player is the fledgling Tata Information Systems Ltd (TISL), which exported Rs 25 crore of hardware in 1993-94 to parent IBM. However, TISL itself hasn't set up a manufacturing facility—yet—and its exports are only a sourcing operation for IBM. In fact, the real growth in hardware exports appears to have come from companies which run export-only projects out of the export processing zones (EPZs) and electronic hardware technology parks (EHTPs).

Just three exporters—the Rs 130.7-crore, PCL Group company, Altos India, the Rs 215.82-crore Tandon Computers, and the Rs 55.12-crore Silicon Valley Technology (SVT)—accounted for 70 per cent of the country's hardware exports last year. While Altos and SVT have set up fully-automated assembly plants that manufacture printed circuit boards (PCBs)—using surface mounting technology (SMT)—for PCs, Tandon has specialised in manufacturing highly labour-intensive computer parts here.

What emerges, as a corollary, from the new hardware boom is that a plant which assembles state-of-the-art PCBs can operate as viably in India as in any other country in the Pacific Rim. For, the duty-free imports that the units in the special zones are allowed—and the low cost of manpower—ensures that India is as competitive as Taiwan or Singapore for manufacturing hardware. Of course, costs are only a part of the story; the would-be Indian exporter has to put in twice the effort in managing his marketing, his quality, and the logistics to earn his place in the $17-billion contract

SANJAY PANDYA



M.L. TANDON

CEO, Tandon Group, Tandon Computer

Sales: Rs 216 crore; **Exports:** Rs 201 crore; **Investment:** Rs 44 crore

THE TANDON Group, based in the Santa Cruz Electronics EPZ (Bombay), is actually the third largest computer hardware manufacturer in India, behind only the Rs 340-crore HCL-HP and the Rs 337-crore Wipro Infotech. And with exports of Rs 201 crore in 1993-94, Tandon is the country's largest hardware exporter. Tandon's success has been built around a systematic design of its business to leverage the advantages that operating out of this country offer. Says CEO Manohar Lal Tandon: "The only factor that is massively in India's favour is low labour costs. And we have always manufactured only those products that have very high manual inputs." A good 95 per cent of Tandon's turnover comes from power supplies for PCs, and certain sub-assemblies that go into hard disk drives, which have labour-added values of over 20 per cent. Since labour costs in Singapore and Taiwan are higher than in India, even after factoring in higher productivity, Tandon chose to produce these parts here.

The other problem—the acceptability of an Indian supplier—was solved by the US-based Tandon Computers owned by M.L. Tandon's brother, Juggi Tandon. And Tandon now sells power supplies to players like IBM and Compaq, and hard disk sub-assemblies to Seagate, Western Digital, and Conner. In fact, one in every 10 PCs sold in the world today has a Tandon power supply. Tandon's challenge today is to maintain the quality produced by its 5,500-strong workforce. All-manual work processes mean that quality is dependent on an individual worker. And Tandon identifies its training systems as the key factor. "With many processes handled by semi-skilled workers, quality is dependent on the effectiveness of our training systems," he says.